

CROP PROFITABILITY & ROI ANALYSIS

District-wise Analysis of Kharif Crops Using Government Production & MSP Data (2024-25)

A Power BI Dashboard Project | Built with Public Data from data.gov.in



Figure 1: National Overview — KPI cards, Profit by Crop, Profit by State, and full district-level detail table with State/Crop slicers

This is the landing page of the dashboard. The four cards at the top summarize the entire dataset in one glance — total Cost, Revenue, Profit, and average ROI% across all 700+ districts and 8 Kharif crops. The two bar charts break this down by Crop and by State, while the table on the right lets a viewer scroll through every individual district-crop combination. The Crop and State panels on the right act as slicers — clicking any crop or state filters every visual on the page instantly.

Project Snapshot

700+	8	29	23,282
Districts Covered	Kharif Crops Analyzed	States/UTs	Data Rows Processed

1. Introduction & Objective

India's agricultural policy leans heavily on the Minimum Support Price (MSP) mechanism to guarantee farmers a fair return on their produce. But how much of a real margin does MSP actually deliver once the government's own declared Cost of Production is factored in — and how does that margin vary by crop, by state, and by district?

This project set out to answer that question using two public datasets: district-wise crop production figures for 2024-25, and the government's official Cost of Production vs MSP data for major Kharif crops. The goal was to build an interactive Power BI dashboard that surfaces Revenue, Cost, Profit, and Return on Investment (ROI) at a granular, district level — not just national averages.

Data Sources

- District-wise Area, Production & Yield (2024-25) — Directorate of Economics & Statistics (DES), Ministry of Agriculture
- Kharif Crop-wise Cost of Production, MSP & Margin (2021-22 to 2024-25) — Rajya Sabha Starred Question No. 58, Session 266 (data.gov.in)

Scope

The analysis covers 8 Kharif crops for which both production data and official Cost/MSP figures were available: Rice, Jowar, Bajra, Ragi, Maize, Tur (Arhar), Moong, and Urad. Rabi crops (Wheat, Gram, Barley, Lentil) were excluded from the financial analysis as their MSP data was not part of this dataset.

2. Data Cleaning & Methodology

Cleaning Steps Applied

- Corrected column data types — Area, Production, and Yield fields converted to proper numeric (decimal) types
- Handled missing values — 28 rows with null Production/Yield (all corresponding to near-zero Area entries) were filled with 0 rather than dropped, to preserve district coverage
- Removed exact duplicate rows across the full dataset
- Trimmed whitespace and standardized text casing in State, District, Crop, and Season fields
- Converted 'NA' placeholder text in the MSP dataset to true nulls, then set Cost/MSP/Difference columns to Whole Number type
- Standardized crop naming (e.g. 'Paddy (Common)' → 'Rice', 'Arhar (Tur)' → 'Tur') to enable a clean join between the two source tables

Key Methodology Notes

- Filtered to Season = 'Total' rows only — the raw data includes Kharif, Rabi, Summer and Total rows per crop-district; using Total avoids double-counting production that would occur if all season rows were summed
- Production was converted from Lakh Tonnes to Quintals (the unit MSP/Cost figures are quoted in) using: 1 Lakh Tonne = 10,00,000 Quintals
- Revenue = Production (Quintals) × MSP per Quintal; Cost = Production (Quintals) × Cost of Production per Quintal; Profit = Revenue – Cost; ROI% = Profit ÷ Cost
- Kharif-season MSP was applied to each crop's full annual (Total) production. Where a crop also has meaningful Rabi/Summer output, this is an approximation, not an exact figure — flagged here for transparency

Tools Used

- Power Query (Power BI) — data cleaning, type correction, unpivoting
- DAX Measures — Revenue, Cost, Profit, and ROI% calculated dynamically via SUMX and RELATED functions across a relational model
- Power BI Desktop — 3-page interactive report with cards, bar charts, slicers, and drill-through tables

3. Key Findings

3.1 Cost, Revenue & ROI by Crop

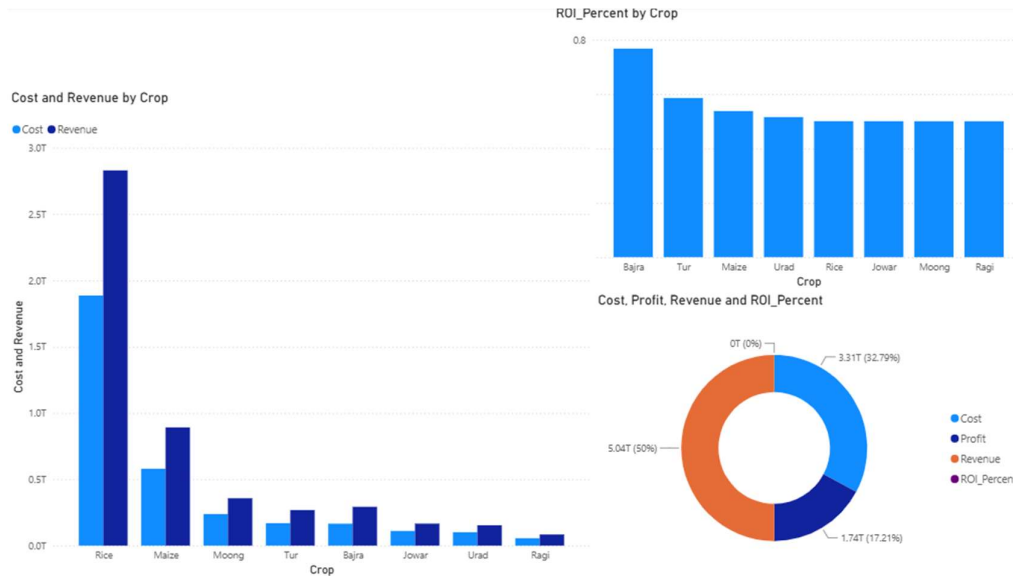


Figure 2: Cost vs Revenue by Crop, ROI% by Crop, and overall Cost/Profit/Revenue split

The left chart compares each crop's total Cost against its total Revenue — the gap between the two bars is the absolute profit. Rice shows the widest gap in absolute terms simply because it is grown at massive scale. The top-right chart tells a different story: it ranks crops by ROI% (profit as a percentage of cost), and here Bajra comes out on top at roughly 78%, followed by Tur at around 59% — meaning these two crops return the most profit per rupee spent on production, even though their total volumes are much smaller than Rice's. The donut chart puts the national totals in perspective — of the total ₹5.04T in revenue generated across all 8 crops, ₹3.31T (66%) goes toward covering cost of production, leaving ₹1.74T (34%) as aggregate profit.

Crop	Observation
Rice	Highest absolute profit — largest production base
Maize	Second-highest profit; strong across all 3 seasons
Bajra	Highest ROI% (~78%) — most efficient margin
Ragi & Jowar	Lowest absolute profit — limited geographic spread

3.2 District-level Drill-down

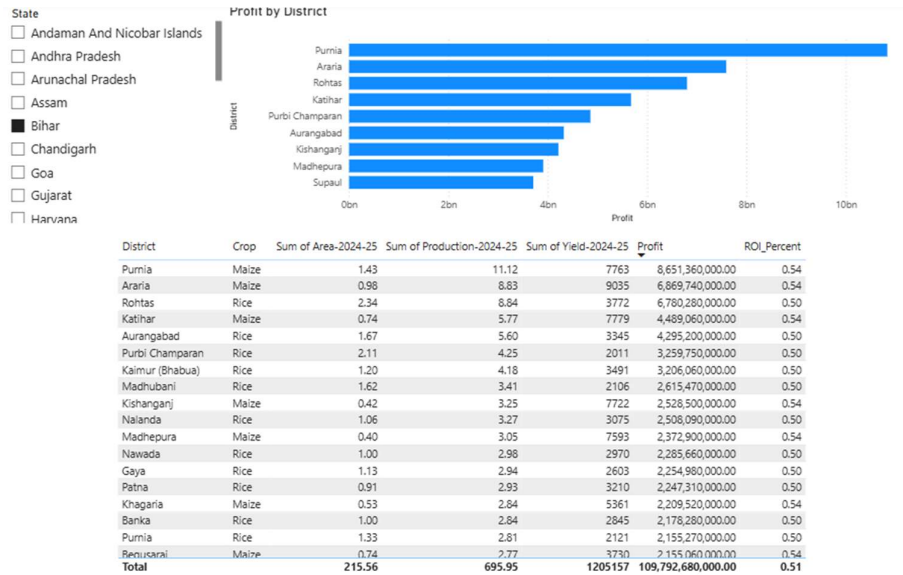


Figure 3: District-level Profit ranking within a selected state (Bihar shown), with full district-crop detail table

This view is filtered to Bihar using the state slicer, and it illustrates how the dashboard supports drill-down analysis for any individual state. Within Bihar, Purnia stands out with over ₹8.6bn in profit from Maize alone — more than double the next-highest district, Araria. The pattern across most districts leans toward Rice and Maize as the dominant profitable crops, with ROI% holding steady in the 0.50–0.54 range regardless of district, which suggests that at the state level, the profit differences are driven mainly by production volume rather than by any district having a fundamentally better cost structure.

This same view can be repeated for any of the 29 states/UTs simply by changing the slicer selection — making it possible to identify the top-performing districts anywhere in the country without rebuilding the report.

4. Conclusion & Takeaways

- MSP does deliver a positive margin over the government's own declared cost of production across all 8 Kharif crops studied — no crop showed a loss at the aggregate level
- Total profit and ROI% tell different stories: Rice leads on scale, Bajra leads on efficiency. Policy or investment decisions should look at both metrics, not just one
- Profit is geographically concentrated — a relatively small set of high-yield districts (e.g. Purnia, Narmadapuram, Nashik) drive a large share of the total national profit figure
- This analysis is an approximation built on public MSP/Cost data for Kharif crops only; Rabi crops (Wheat, Gram, Barley, Lentil) and crop-specific input cost variation by region are natural next steps

Dashboard Structure

Page	Contents
1. Crop ROI Analysis	National KPIs (Revenue, Cost, Profit, ROI%), Profit by Crop, Profit by State, detail table
2. District-level Drill Down	State slicer, top districts by profit, full district-crop detail table
3. Cost vs MSP Analysis	Cost vs Revenue comparison by crop, ROI% by crop, Cost/Profit/Revenue donut

Data Sources

Directorate of Economics & Statistics (DES), Ministry of Agriculture & Farmers Welfare — District-wise Area, Production & Yield 2024-25.

Rajya Sabha Starred Question No. 58, Session 266 — Cost of Production, MSP and Margin, Kharif Crops (data.gov.in):
<https://www.data.gov.in/resource/kharif-crop-wise-details-minimum-support-price-msp-and-cost-production-2021-22-2024-25>

Rajya Sabha Starred Question No. 58, Session 266 — Cost of Production, MSP and Margin, Commercial Crops (data.gov.in):
<https://www.data.gov.in/resource/commercial-crop-wise-details-minimum-support-price-msp-and-cost-production-2021-2024>

MSP Statement, Department of Agriculture & Farmers Welfare: <https://desagri.gov.in/statistics-type/latest-minimum-support-price-msp-statement/>